Laeeq Aslam

Machine Learning Engineer | AI & Edge Computing | Sustainable Systems
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Professional Summary

Machine learning researcher with a PhD in Control Science and Machine Learning, specializing in **physics-informed models** and **AI for sustainable systems**. Experienced in **time-series forecasting**, **edge computing**, and **optimization algorithms**. Eager to advance cutting-edge research in AI and renewable energy systems, with a strong record of first-author publications and interdisciplinary collaborations. Seeking to contribute to impactful AI-driven solutions in **energy systems** and **climate-related challenges**.

Publications & Research

Published **8 journal** articles, **4 arXiv preprints**, **3 conference papers,** and **3 under review,** with 31 total citations. Recent publications as **first author** are as follows,

- ***Physics-Informed Spatio-Temporal Network with Trainable Adaptive Feature Selection for Short-Term Wind Speed Prediction (Computers & Electrical Engineering, 2025).*** *Developed a novel spatio-temporal model for wind speed prediction, achieving more than 9.5% improvement in accuracy compared to state-of-the-art models, showcasing the integration of physics-based knowledge in machine learning.*
- **Dynamic Optimization of Recurrent Networks for Wind Prediction on Edge Devices. *(IEEE Access, 2025).*** *Proposed an optimization algorithm, "adaptive Simulated Annealing with Memory-based Rejection (aSAR)" that optimizes hyperparameters for wind speed prediction models, incorporating dynamic cooling and memory rejection to prevent revisiting suboptimal solutions. This approach reduces model size by up to 98.75% while enhancing prediction accuracy by up to 54.17% compared to traditional methods.*
- **Hardware-Centric Exploration of the Discrete Design Space in Transformer–LSTM Models for Wind Speed Prediction on Memory-Constrained Devices. *(Energies,2025).*** *This study integrates hardware considerations into the optimization process for a hybrid transformer-LSTM model, with a focus on achieving real-time performance. Models are trained on a server, pushed to hardware via gRPC over a local network, and tested on the Jetson Nano device. This approach ensures that hyperparameters are optimized while considering the real-time performance of the model on edge devices, such as the Jetson Nano.*
- **Integrating Physics-Informed Vectors for Improved Wind Speed Forecasting with Neural Networks. *(Asian Control Conference, 2024).*** *This research improves wind speed forecasting by integrating Physics-Informed Vectors (PIV) into LSTM and TCN models. The approach uses atmospheric data from multiple directions to calculate PIV, which improves model accuracy. A custom hybrid loss function accelerates model convergence. The method shows significant improvements in prediction performance across datasets from multiple countries.*

Under Review

- ***Wind Speed Prediction Using a Dynamic Tree-structured Parzen Estimator Optimized Shallow Hybrid Model. (Major Revision Completed and Resubmitted, Computers & Electrical Engineering).*** *This work proposes the D-TPE-SHM model for wind speed prediction on memory-constrained devices, combining FICA, Adaptive LASSO, LSTM, and TCN to capture temporal patterns, with hyperparameters optimized using the dynamic D-TPE algorithm, which adjusts thresholds based on gradient feedback to improve convergence speed and model performance. The model outperforms existing methods in accuracy and efficiency, making it suitable for real-time deployment on devices such as Jetson Nano.*
- **A Distribution Matching Framework for Zero-Shot Wind Speed Prediction on Edge Devices in Unseen Locations.** ***(Submitted, Engineering Applications of Artificial Intelligence).*** *This work proposes a Distribution Matching Framework (DMF) for zero-shot wind speed prediction on resource-constrained edge devices, utilizing pre-trained location-specific recurrent neural networks with attention (RNNA) across 460 locations. By employing Empirical Distribution Matching (EDM) and Multi-Objective Loss Optimization (MOLO), the framework ensures efficient model deployment for small-scale wind turbines, achieving robust performance with a compact model size of 7.3 MB. The method demonstrates superior performance with low Jensen-Shannon divergence (below 0.0015) across five unseen locations.*
- **Perception-Informed Neural Network** for Highly Volatile Wind Speed Prediction. ***(In Write-Up Phase).***

Work Experience

Bond and Built Pvt Ltd, Pakistan

AI/ML Consultant - Edge Computing & Computer Vision | July 2024 – March 2025

- Led the development of a real-time **footwear analytics system** deployed across urban Pakistan.
- Scaled system to process **50,000+ daily inferences** with **60+ concurrent streams**.
- Delivered actionable **market intelligence** for strategic material investments.

Central South University, China

PhD Researcher – Machine Learning & AI for Sustainable Systems | Sept 2020 – Present

- Developed **Physics-Informed Spatio-Temporal Network (PISTNet)**, achieving **more than 9.5% improvement in SMAPE** against state-of-the-art models, including iTransformers, PatchTST, and Mixformer for wind speed forecasting.
- Proposed **Adaptive Simulated Annealing with Memory-based Rejection for Recurrent Neural Networks Discrete Hyper-Parameter Space optimization**, reducing **MAE by 14.5%** against baseline models (LSTM, TCN) in time-series forecasting.
- Contributed to the Key **R&D Program of Hunan Province, Project 2020WK2007**, focusing on optimizing energy integration and improving grid performance for wind energy storage.

DLISION, Pakistan

Machine Learning Engineer – Computer Vision & AI Deployment | May 2021 – Sept 2022

- Led **computer vision projects** focused on **semantic segmentation and real-time object detection**.
- Optimized **CNN-UNet & Swin UNet models**, improving **segmentation accuracy by 3% against their baseline architectures** using **Focal Loss**.

International Islamic University, Pakistan

Higher Education Research Assistant – AI for Healthcare | March 2019 – Feb 2020

- Applied **Generative Adversarial Networks (GANs)** for **breast cancer detection**, improving classification accuracy.
- Conducted data augmentation techniques that **increased dataset diversity** and **enhanced model robustness**.

iUSE School of Engineering and Management Sciences, Pakistan

Lecturer & Lab Engineer – Embedded Systems & AI | Sept 2013 – Feb 2019

- Taking undergraduate courses related to **Embedded Systems**, **Programming,** and **Discrete Signal Processing**.
- Designed and developed an **Instrumentation & Measurement Lab** for **sensor-based data acquisition in MATLAB**.
- Led **student research projects** on **energy management, accident alerts, and secure electronic voting**.

Education

- PhD in Control Science & Machine Learning | Central South University, China | 2020 – Present.
- MS in Electronic Engineering| International Islamic University, Pakistan | 2015 – 2017

Technical Skills

- **Machine Learning & AI:** Deep Learning | Computer Vision | Time-Series Analysis | Predictive Modeling.
- **Programming & Tools:** Python | MATLAB | C++ | TensorFlow | PyTorch | Keras | OpenCV | Edge Impulse.
- **DevOps & Deployment:** Docker | AWS (E2C)| Git | Triton Inference Server.
- **Hardware & IoT:** Raspberry Pi | NVIDIA Jetson | Arduino Nano BLE Sense.
- **Data Analysis & Optimization:** Pandas | NumPy | Matplotlib | Seaborn | Feature Engineering

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Key Projects & Research Contributions

- Deployed a **real-time edge-to-cloud footwear analytics pipeline** (YOLOv5 → DenseNet-121), processing **50k+ daily inferences** across urban Pakistan with **60+ concurrent streams.**
- **AI-Based Sports Analytics:** Enhanced **real-time player segmentation** using **U-NET models**, improving **inference time by 15%**.
- Developed and published the **keras-swin-unet** PIP package for satellite imagery segmentation and the **TimeMesh** library for efficient time series data preprocessing.

Awards & Honors

- Gold Medalist – MS in Electronic Engineering | International Islamic University.
- Hunan Provincial Government Funding for Research & Development (**Project 2020WK2007**).
- Chinese Government CSC Scholarship for PhD Studies.